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Mathematical Analysis

1st Year Computer Science

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※ Real numbers

Let us start with some standard notation: $\emptyset$ is the empty set; $\mathbb{N} = \{1, 2, \dots\}$ the set of natural numbers; $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\} = \{m - n \mid m, n \in \mathbb{N}\}$ the set of integers; $\mathbb{Q} = \{\frac{m}{n} \mid m, n \in \mathbb{Z}, n \neq 0\}$ the set of rational numbers; $\mathbb{R}$ the set of real numbers.

You are very much used with the real numbers. However, it is not trivial to define them in a rigorous way, see for example [1] or [2] for different methods of constructing $\mathbb{R}$ from $\mathbb{Q}$. We will straightforwardly start working with the real numbers – for this reason some of their properties, e.g. Definition 1.5, will simply be given as definitions.

Definition 1.1

Let A be a subset of $\mathbb{R}$, denoted as $A \subseteq \mathbb{R}$. We define $x \in \mathbb{R}$ to be

a *lower bound* for A if $x \leq a, \forall a \in A$; an *upper bound* for A if $x \geq a, \forall a \in A$.

We define

$lb(A) := \{x \in \mathbb{R} \mid x \leq a, \forall a \in A\}$ the set of lower bounds of A ,

$ub(A) := \{x \in \mathbb{R} \mid x \geq a, \forall a \in A\}$ the set of upper bounds of A .

We define $x \in \mathbb{R}$ to be

the *minimum* of A if $x \in lb(A) \cap A$; the *maximum* of A if $x \in ub(A) \cap A$,

denoted by $\min(A)$, respectively $\max(A)$. In other words, we have that

$\min(A) \in A$ and $\min(A) \leq a, \forall a \in A$; $\max(A) \in A$ and $\max(A) \geq a, \forall a \in A$.

Note that there are sets which do not have minimum or maximum, e.g. $(0, 1)$.

Definition 1.2

A set $A \subseteq \mathbb{R}$ is defined to be

- bounded (from) below if $lb(A) \neq \emptyset$;
- bounded (from) above if $ub(A) \neq \emptyset$;
- bounded if it is both bounded below and above;
- unbounded if it is not bounded.

Definition 1.3

We say that $x \in \mathbb{R}$ is the *supremum* of $A \subseteq \mathbb{R}$, $x := \sup(A)$, if and only if:

1. $x \geq a, \forall a \in A$, that is $x \in ub(A)$.
2. if u is an upper bound for A , then $x \leq u$.

The supremum is *the least upper bound*, i.e. $\sup(A) := \min(ub(A))$.

Definition 1.4

We say that $x \in \mathbb{R}$ is the *infimum* of $A \subseteq \mathbb{R}$, $x := \inf(A)$, if and only if:

1. $x \leq a, \forall a \in A$, that is $x \in lb(A)$.
2. if u is a lower bound for A , then $x \geq u$.

The infimum is *the greatest lower bound*, i.e. $\inf(A) := \max(lb(A))$.

Definition 1.5 (Completeness Axiom)

Every set $A \subseteq \mathbb{R}$ that is bounded above has a supremum. Similarly, every set $A \subseteq \mathbb{R}$ that is bounded below has an infimum.

Note that if A has a maximum, then $\sup(A) = \max(A)$. Similarly, if A has a minimum, then $\inf(A) = \min(A)$. Also, if $\sup(A) \in A$, then $\max(A) = \sup(A)$.

Example 1.6

(a) $A = \{\frac{1}{n} \mid n \in \mathbb{N}\}$, $\sup(A) = 1 = \max(A)$, $\inf(A) = 0$, $\nexists \min(A)$.

(b) $A = \{x \in \mathbb{Q} \mid x^2 \leq 2\}$, $\sup(A) = \sqrt{2}$, $\nexists \max(A)$, $\inf(A) = -\sqrt{2}$, $\nexists \min(A)$.

Proposition 1.7

Let $A \subseteq \mathbb{R}$ be a bounded set. For $\sup(A)$ and $\inf(A)$ the following are true:

$$\forall \varepsilon > 0, \exists x \in A \text{ such that } \sup(A) - \varepsilon < x,$$

$$\forall \varepsilon > 0, \exists x \in A \text{ such that } x < \inf(A) + \varepsilon.$$

Proof. By definition, for any $y < \sup(A)$, say $y = \sup(A) - \varepsilon$ with $\varepsilon > 0$, we have that $y \notin ub(A)$. Hence there exists $x \in A$ such that $y < x$. Similar proof for $\inf(A)$. $\square$

Proposition 1.8

Let $A \subseteq B \subseteq \mathbb{R}$ be (nonempty) bounded sets. Then

$$\inf(B) \leq \inf(A) \leq \sup(A) \leq \sup(B)$$

and

$$\sup(A \cup B) = \max\{\sup(A), \sup(B)\},$$

$$\inf(A \cup B) = \min\{\inf(A), \inf(B)\}.$$

Proof. It follows directly from the definitions. □

Definition 1.9

Define the *extended real line* $\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, \infty\}$, where ∞ and $-\infty$ are such that

$$\forall x \in \mathbb{R}, -\infty < x < \infty.$$

If a set A is not bounded above, we define $\sup(A) := \infty$.

If a set A is not bounded below, we define $\inf(A) := -\infty$.

[Seminar] The empty set $\emptyset$ is bounded by any number. In $\overline{\mathbb{R}}$, $\sup(\emptyset) = -\infty$, $\inf(\emptyset) = \infty$.

Definition 1.10

A set $V \subseteq \mathbb{R}$ is a *neighborhood (vecinity)* of $x \in \mathbb{R}$ if

$$\exists \varepsilon > 0 \text{ such that } (x - \varepsilon, x + \varepsilon) \subseteq V.$$

A set $V \subseteq \mathbb{R}$ is a *neighborhood* of ∞ if $\exists a \in \mathbb{R}$ such that $(a, \infty) \subseteq V$.

A set $V \subseteq \mathbb{R}$ is a *neighborhood* of $-\infty$ if $\exists a \in \mathbb{R}$ such that $(-\infty, a) \subseteq V$.

We denote all the neighborhoods of x by

$$\mathcal{V}(x) := \{V \subseteq \mathbb{R} \mid V \text{ is a neighborhood of } x\}.$$

Definition 1.11

Let $A \subseteq \mathbb{R}$. The following set is called the *interior* of A

$$\text{int}(A) := \{x \in \mathbb{R} \mid \exists V \in \mathcal{V}(x) \text{ such that } V \subseteq A\},$$

and the following set is called the *closure* of A

$$\text{cl}(A) := \{x \in \mathbb{R} \mid \forall V \in \mathcal{V}(x), V \cap A \neq \emptyset\}.$$

Proposition 1.12

For any $A \subseteq \mathbb{R}$, it holds that $\text{int}(A) \subseteq A \subseteq \text{cl}(A)$.

Proof. To prove that $\text{int}(A) \subseteq A$ we prove that if $x \in \text{int}(A)$, then $x \in A$. Let $x \in \text{int}(A)$, then $\exists \varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \subseteq A$. Since $x \in (x - \varepsilon, x + \varepsilon)$, we have that $x \in A$.

To prove that $A \subseteq \text{cl}(A)$ we show that if $x \in A$, then $x \in \text{cl}(A)$. Let $x \in A$. Then for any $V \in \mathcal{V}(x)$ it holds that $x \in V$, giving that $x \in V \cap A$. Hence $x \in \text{cl}(A)$ since $V \cap A \neq \emptyset$. $\square$

Definition 1.13

If $A = \text{int}(A)$, then A is called *open*. If $A = \text{cl}(A)$, then A is called *closed*.

Remark 1.14

To prove that a set A is open, it is sufficient to prove that $A \subseteq \text{int}(A)$.

To prove that a set A is closed, it is sufficient to prove that $\text{cl}(A) \subseteq A$.

Proposition 1.15

The following statements are true:

- The complement of an open set is closed.
- The complement of a closed set is open.

Proof. Let us prove the first statement, the other one being similar. Consider A an open set, i.e. $A = \text{int}(A)$, and denote by $A^c = \{x \in \mathbb{R} \mid x \notin A\}$ its complement. To prove that A^c is closed, we prove that $\text{cl}(A^c) \subseteq A^c$. Consider $x \in \text{cl}(A^c)$ and let's assume that $x \notin A^c$, i.e. $x \in A$, aiming to obtain a contradiction. Since A is open, there exists $V \in \mathcal{V}(x)$ such that $V \subseteq A$, giving that $V \cap A^c = \emptyset$: contradiction with $x \in \text{cl}(A^c)$. Hence the assumption $x \notin A^c$ is false, and we have that if $x \in \text{cl}(A^c)$, then $x \in A^c$. In other words, $\text{cl}(A^c) \subseteq A^c$. $\square$

Proposition 1.16

Any union of open sets is open. Any intersection of closed sets is closed. Any finite intersection of open sets is open. Any finite union of closed sets is closed.

Proof. (Optional) Left as extra homework. $\square$

Exercises. Seminar 1

1. Find the lower and the upper bounds, then $\sup$, $\inf$, $\max$, $\min$ for each of the following:

- | | |
|----------------------------------|---------------------------------|
| (a) $[-3, 2) \cup \{3\}$. | (c) $(-5, 5) \cap \mathbb{Z}$. |
| (b) $(-1, 1] \cup (2, \infty)$. | (d) $\emptyset$. |

2. Find the $\sup$, $\inf$, $\max$, $\min$ for each of the following sets:

- | | |
|--|--|
| (a) $\{x \in \mathbb{Q} \mid x^2 < 3\}$. | (c) $\{\frac{n}{n+1} \mid n \in \mathbb{N}\}$. |
| (b) $\{x^2 - 4x + 3 \mid x \in \mathbb{R}\}$. | (d) $\{2^{-k} + 3^{-m} \mid k, m \in \mathbb{N}\}$. |

3. Suppose that S is nonempty and bounded above. Show that the set $-S := \{-x \mid x \in S\}$ is bounded below and $\inf(-S) = -\sup(S)$.

4. Let $f : D \rightarrow \mathbb{R}$ and $g : D \rightarrow \mathbb{R}$ be two functions defined on a nonempty set D . Prove that

$$\inf_{x \in D} (f(x) + g(x)) \geq \inf_{x \in D} f(x) + \inf_{x \in D} g(x) \quad \text{and} \quad \sup_{x \in D} (f(x) + g(x)) \leq \sup_{x \in D} f(x) + \sup_{x \in D} g(x).$$

Give examples where the above inequalities are strict.

5. Which of the following sets are neighborhoods of 0?

$$[-1, 1] \cup \{2\}; \quad (-1, 1) \cap \mathbb{Q}; \quad \bigcap_{n=1}^{\infty} \left[-\frac{1}{n}, \frac{1}{n}\right].$$

6. Let $x \in \mathbb{R}$ and $U, V \in \mathcal{V}(x)$. Prove that $U \cap V \in \mathcal{V}(x)$.

7. Find the interior and the closure for each of the following sets:

(a) $(1, 2]$.

(c) $(-1, 1] \cup (2, \infty)$.

(b) $[-3, 2) \cup \{3\}$.

(d) $(-5, 5) \cap \mathbb{Z}$.

Exercises. Self study

1. Let $a, b \in \mathbb{R}$ with $a > 0$. If $S \subset \mathbb{R}$ is nonempty and bounded above, prove that

$$\sup_{x \in S} (ax + b) = a \sup(S) + b.$$

2. Let $a, b \in \mathbb{R}$. Prove that there exist neighborhoods $U \in \mathcal{V}(a)$ and $V \in \mathcal{V}(b)$ such that

$$U \cap V = \emptyset.$$

3. Let $A = (0, 1) \cap \mathbb{Q}$. Show rigorously (using the definitions) that

$$\inf A = 0, \sup A = 1, \text{int} A = \emptyset \text{ and } \text{cl} A = [0, 1].$$

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